Cellular Respiration Outline

Honors High School Biology

Cellular respiration transfers chemical energy from glucose into ATP that cells can use. The pathway includes glycolysis, pyruvate oxidation, the Krebs cycle, and the electron transport chain.

I. Cellular Respiration Overview

- Overall function: release energy from glucose and capture part of that energy in ATP.
- Overall equation: glucose + oxygen → carbon dioxide + water + ATP.
- Glucose becomes oxidized as it loses electrons. Oxygen becomes reduced when it accepts electrons at the end of the pathway.
- NAD+ and FAD collect high-energy electrons and hydrogen ions, forming NADH and FADH2.
- Aerobic respiration requires oxygen and produces much more ATP than fermentation.

II. Glycolysis

- Location: Cytosol
- Main purpose: Split one six-carbon glucose into two three-carbon pyruvate molecules and capture a small amount of energy.
- Inputs
  - 1 glucose
  - 2 ATP invested
  - 2 NAD+ and ADP + phosphate
- Major events
  - The energy-investment phase uses two ATP to activate glucose.
  - The six-carbon molecule separates into two three-carbon molecules.
  - Oxidation transfers electrons to NAD+, producing NADH.
  - Substrate-level phosphorylation transfers phosphate directly to ADP.
- Outputs per glucose
  - 2 pyruvate
  - 2 NADH
  - 4 ATP produced, with a net gain of 2 ATP
  - Water
- Oxygen is not used directly during glycolysis. Without oxygen, glycolysis can continue only if fermentation regenerates NAD+.

III. Pyruvate Oxidation

- Location: Mitochondrial matrix
- Main purpose: Prepare pyruvate to enter the Krebs cycle by converting it into acetyl-CoA.
- Inputs
  - 2 pyruvate
  - 2 NAD+
  - 2 coenzyme A
- Major events
  - One carbon is removed from each pyruvate and released as carbon dioxide.
  - Electrons reduce NAD+ to NADH.
  - Coenzyme A attaches to each remaining two-carbon acetyl group.
- Outputs per glucose
  - 2 acetyl-CoA
  - 2 NADH
  - 2 CO_2
  - No ATP

- Oxygen is required indirectly: without a functioning ETC, NAD+ is not replenished and pyruvate oxidation cannot continue.

IV. Krebs Cycle

- Location: Mitochondrial matrix
- Main purpose: Complete the oxidation of glucose-derived carbon and load electron carriers for the ETC.
- Inputs
  - 2 acetyl-CoA
  - NAD+
  - FAD
  - ADP + phosphate
- Major events
  - Each acetyl group combines with four-carbon oxaloacetate to form six-carbon citrate.
  - Citrate undergoes oxidation and decarboxylation reactions.
  - NAD+ and FAD accept electrons, forming NADH and FADH2.
  - Oxaloacetate is regenerated, allowing the cycle to continue.
  - The cycle turns twice for each glucose because glycolysis produces two pyruvates.
- Outputs per glucose
  - 6 NADH
  - 2 FADH2
  - This cycle also depends indirectly on oxygen because the ETC must regenerate NAD+ and FAD for the cycle to continue.
  - 2 ATP
  - 4 CO_2

V. Electron Transport Chain and Oxidative Phosphorylation

- Location: Inner mitochondrial membrane, especially the cristae
- Main purpose: Use energy from electrons to build a proton gradient and produce most of the cell's ATP.
- Inputs
  - 10 NADH
  - 2 FADH2
  - Oxygen
  - ADP + phosphate
- Major events
  - NADH and FADH2 donate electrons and regenerate NAD+ and FAD.
  - Electrons move through membrane protein complexes and release energy.
  - The complexes use this energy to pump H^+ from the matrix into the intermembrane space.
  - The resulting electrochemical gradient stores potential energy.
  - H^+ diffuses through ATP synthase, driving ATP production through chemiosmosis.
  - Oxygen serves as the final electron acceptor and combines with electrons and H^+ to form water.
- Outputs per glucose
  - Approximately 32 to 34 ATP using the convention in this course
  - Water
  - Regenerated NAD+ and FAD

VI. ATP Production Mechanisms

- Substrate-level phosphorylation: an enzyme transfers a phosphate directly from a molecule to ADP.
  - Occurs during glycolysis and the Krebs cycle.
- Oxidative phosphorylation: the ETC builds a proton gradient that powers ATP synthase.
  - Produces most of the ATP during aerobic respiration.

VII. Overall Products Per Glucose

- Carbon dioxide: 6 total, with 2 from pyruvate oxidation and 4 from the Krebs cycle.
- Electron carriers delivered to the ETC: 10 NADH and 2 FADH2.
- ATP: approximately 36 to 38 total using the convention in this course.
- Water forms when oxygen accepts electrons and hydrogen ions at the end of the ETC.
- ATP totals can vary because cells differ in how efficiently they transfer and use energy.

VIII. When Oxygen Is Unavailable

- Without oxygen as the final electron acceptor, the ETC stops and can no longer regenerate NAD+ and FAD from NADH and FADH2.
- As NAD+ and FAD become unavailable, pyruvate oxidation and the Krebs cycle also stop. Thus, the aerobic stages after glycolysis cannot continue without oxygen.
- Fermentation regenerates NAD+ so glycolysis can continue.
  - Lactic acid fermentation converts pyruvate to lactate in animal cells and some bacteria.
  - Alcoholic fermentation produces ethanol and carbon dioxide in yeast and some microorganisms.
- Fermentation adds no ATP beyond the 2 net ATP produced by glycolysis.

IX. Essential Connections

- Carbon flow: glucose → pyruvate → acetyl-CoA → carbon dioxide.
- Electron flow: glucose-derived molecules → NADH and FADH2 → ETC → oxygen.
- Energy flow: glucose bonds → electron carriers → proton gradient → ATP.
- Oxygen's role: accept electrons at the end of the ETC so electron flow and ATP production can continue.
- Carbon dioxide's role: waste product released when carbon-containing molecules are oxidized.

X. Major Ideas to Remember

- Glycolysis begins respiration in the cytosol and yields two pyruvate and two net ATP.
- Pyruvate oxidation links glycolysis to the Krebs cycle.
- The Krebs cycle produces little ATP directly but generates many electron carriers.
- The ETC and chemiosmosis produce most ATP.
- Oxygen is essential for aerobic respiration because it accepts electrons at the end of the ETC, allowing NAD+ and FAD to be regenerated for pyruvate oxidation and the Krebs cycle.